Deep Research Report

AI-Powered Medical Image Report Generation
Project feasibility, existing systems, architecture, datasets, impact, risks, and recommended scope

Research date: 23 August 2026 | Target audience: 3rd-year undergraduate project team

Executive conclusion

The proposed project is a real and active research area, generally referred to as automatic radiology report generation or broader multimodal medical report generation. Strong systems already exist, including Microsoft's MAIRA family and open research implementations such as R2Gen. Newer medical multimodal models such as Google's MedGemma explicitly support medical image interpretation and report-generation applications. Therefore, the novelty should not be "we generated a report from an image"; it should come from the project's chosen clinical scope, grounding, context fusion, evaluation, explainability, and/or safety architecture. [1-8]

Recommended undergraduate scope: start with chest X-rays, combine the image with a small amount of structured patient context, use a pretrained medical vision/multimodal model, optionally add retrieval from trusted medical sources or prior reports, generate a structured report, and include evidence/uncertainty plus mandatory human review. This is ambitious but realistic without training a foundation model from scratch. [1][3][6]

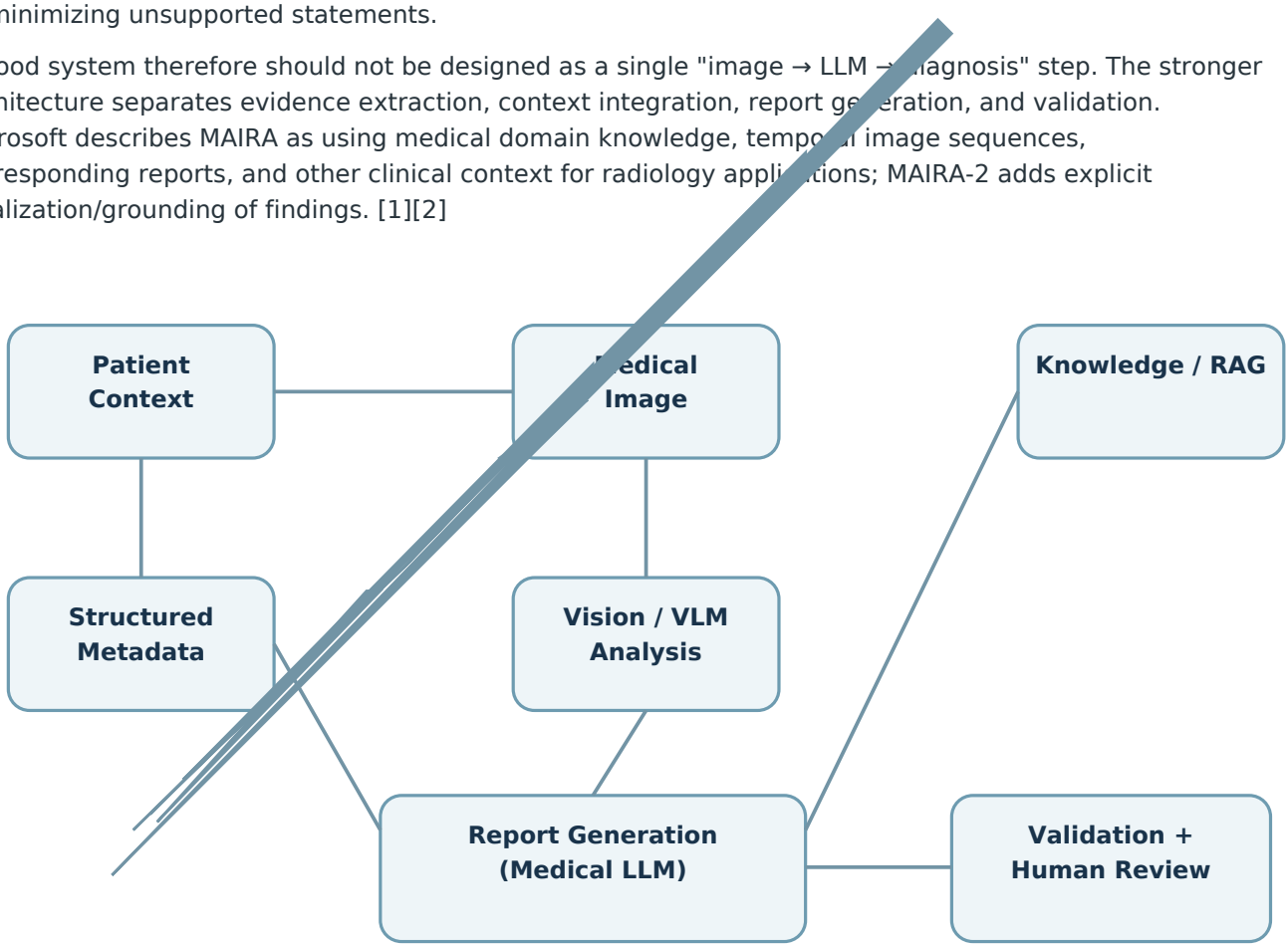
1. What the Project Actually Is

A practical interpretation of the mentor's title is: Design and develop a system that accepts a medical image and relevant clinical context, extracts clinically meaningful findings, and generates a structured, evidence-grounded report for review by a qualified professional.

The project has two distinct AI problems:

- Perception: understand the medical image and identify or describe relevant findings.
- Generation: convert image evidence plus patient context into a coherent, structured report while minimizing unsupported statements.

A good system therefore should not be designed as a single "image → LLM → diagnosis" step. The stronger architecture separates evidence extraction, context integration, report generation, and validation. Microsoft describes MAIRA as using medical domain knowledge, temporal image sequences, corresponding reports, and other clinical context for radiology applications; MAIRA-2 adds explicit localization/grounding of findings. [1][2]



Suggested report sections

Section	Purpose
Clinical history	Symptoms, duration, relevant history and indication for the examination.
Technique / study information	Modality, view, body region and basic study metadata when available.
Findings	Evidence-grounded observations from the image(s).
Impression	Concise synthesis of the important findings.
Recommendations	Only carefully qualified follow-up considerations; not autonomous treatment advice.
Evidence / confidence	Links findings to image regions, extracted labels, retrieved evidence, or uncertainty estimates when available.

2. Functional Requirements

A sensible minimum viable product (MVP) should include the following capabilities.

Requirement	MVP	Stretch goal
User input	Patient age/sex, symptoms, indication, image upload	Prior reports, labs, longitudinal history
Image support	Chest X-ray JPG/PNG	DICOM, multi-view studies, other modalities
Image analysis	Pretrained medical VLM/model	Fine-tuning, localization, uncertainty calibration
Report generation	Structured Findings + Impression	Custom report templates, longitudinal comparison
Grounding	Prompt/schema constraints + evidence display	RAG, image-region grounding, factuality verifier
Evaluation	Clinical label agreement + text metrics	RadGraph/RadCliQ-style factuality, calibration, human review
Security	Authentication, encrypted transport, de-identified data	Audit logs, role-based access, retention policies
Output	Web report + PDF	FHIR-compatible export, EHR integration

Non-functional requirements

- Reliability: the system must avoid inventing findings and clearly represent uncertainty.
- Latency: the demo should feel interactive; model inference and PDF generation should be measurable.
- Reproducibility: fixed dataset splits, model versions, prompts, and evaluation scripts.
- Privacy: use de-identified public data for development and avoid real patient data in a student prototype unless proper approvals and controls exist.
- Human oversight: position the system as decision support/report drafting, not autonomous diagnosis.

3. Does Something Like This Already Exist?

Yes - very much so. The exact idea is an established research field, and several projects are close enough that they should be treated as baselines or reference architectures rather than ignored. [1-5][8-10]

Existing work	What it does	Relevance to your project
Microsoft MAIRA / MAIRA-1	Multimodal radiology research program; MAIRA-1 is a specialized model for radiology report generation.	Very high. Demonstrates the image + clinical context + report-generation direction.
MAIRA-2	Generates chest X-ray reports and introduces grounded report generation, including localization of individual findings and a factuality-oriented evaluation framework.	Extremely high. Good reference for explainability/grounding and evaluation.
RAG chest X-ray report generation	Uses vision-language embeddings to retrieve relevant radiology text, then uses a generative model to create the report; authors explicitly motivate RAG to help control hallucinations.	Very high if you want a RAG component.
R2Gen	Open-source memory-driven transformer for radiology report generation on IU X-Ray and MIMIC-CXR.	High. Useful as a reproducible baseline and code reference.
MedGemma	Open medical models for text and image comprehension; current documentation lists image interpretation and medical image report generation as potential application areas.	Very high for a modern student prototype using a pretrained model.
LLaVA-Med	Biomedical vision-language conversational assistant trained from a general VLM.	Moderate-high. Useful for understanding biomedical VLM adaptation rather than radiology-only reporting.

Implication: your contribution should be framed as a new system combination, adaptation, evaluation, or workflow - not as a claim that automatic medical report generation itself is novel.

Layer	Recommended role	Examples / technologies
Image ingestion	Load, validate, normalize and preview medical images.	PIL/OpenCV; DICOM with pydicom
Medical vision	Extract medically useful image representation/findings.	MedGemma; MAIRA-family research; other pretrained medical VLMs
Medical imaging framework	Training/inference utilities for medical image tasks.	MONAI + PyTorch
Knowledge retrieval	Ground report generation with relevant medical context.	Vector store + trusted medical references/reports
Report generation	Convert evidence into structured clinical text.	Medical-capable multimodal LLM / LLM
Validation	Check factuality/clinical consistency.	CheXbert/RadGraph-style labels; rule checks; verifier model; human review
Backend	Expose AI pipeline to UI.	FastAPI or Flask
Frontend	Upload + report review.	React + Tailwind
Database	Metadata and report storage.	PostgreSQL; object storage
Deployment	Reproducible packaging.	Docker; cloud VM/container

Important: do not build a foundation LLM from scratch. For an undergraduate project, the sensible path is to use an existing pretrained medical model and focus engineering/research effort on the multimodal pipeline, grounding, evaluation, and user workflow.

5. Datasets and Data Strategy

MIMIC-CXR is the most important dataset to investigate first. PhysioNet reports 377,110 chest-radiograph images corresponding to 227,835 studies, with free-text radiology reports; it is de-identified under HIPAA Safe Harbor. Its JPG derivative adds structured labels and recommended train/validation/test splits. Access has requirements, so the team should plan for credentialing and data-use compliance. [11][12]

CheXpert is another major chest X-ray benchmark. Stanford describes 224,316 chest radiographs from 65,240 patients, with labels for 14 observations. It is particularly useful for structured pathology evaluation and classification baselines. [13]

IU X-Ray is smaller and historically popular for report-generation research. Because it is much smaller than MIMIC-CXR, it is useful for fast prototyping but not ideal as the only source for a serious report-generation system. R2Gen, for example, supports both IU X-Ray and MIMIC-CXR. [8][9]

Dataset	Scale / contents	Use in project
MIMIC-CXR / MIMIC-CXR-JPG	377k images; 227k+ studies; reports; de-identified; JPG version provides structured labels.	Primary training/evaluation dataset for report generation.
CheXpert	224k radiographs; 65k+ patients; 14 observations.	External evaluation / pathology classification / robustness.
IU X-Ray	Small paired image-report benchmark.	Fast prototyping and comparison against historical baselines.

Data rule for your prototype

Do not upload real student/family patient data to a cloud LLM for experimentation. Build the system around de-identified public datasets first. If a mentor later wants a hospital pilot, privacy, consent, institutional approval, data retention, and vendor/data-processing terms become separate workstreams.

6. Evaluation: How Do You Prove the Project Works?

A common mistake is to evaluate generated reports only with generic language metrics such as BLEU or ROUGE. A medically plausible sentence can still be clinically wrong. Modern work therefore emphasizes factual and clinical metrics. MAIRA-2 introduced RadFact for sentence-level correctness/completeness and grounded reporting. RadGraph provides structured clinical entities and relations for radiology reports. [2][14]

Evaluation layer	Example metric / test	What it tells you
Text similarity	BLEU, ROUGE, METEOR, BERTScore	How similar the generated text is to a reference report.
Clinical content	CheXbert-derived label agreement; pathology F1/AUROC	Whether important clinical observations match reference labels.
Structured factuality	RadGraph F1 / entity-relation agreement	Whether clinical entities and relations are preserved correctly.
Grounded factuality	RadFact-style sentence-level correctness/completeness	Whether report statements are supported by the image/report evidence.
Uncertainty / calibration	Confidence calibration, ECE, abstention rate	Whether the model knows when it is uncertain.
Human evaluation	Blind review by qualified clinicians or trained evaluators	Clinical usefulness, readability, omissions and harmful errors.

Best undergraduate experiment: compare three pipelines on the same test set: (A) base multimodal model, (B) model + structured patient context, and (C) model + context + retrieval/grounding. Report both language and clinical factuality metrics. That creates a clean research question and measurable contribution.

7. Real-World Impact

The potential impact is substantial because radiology reporting is a high-volume information-generation task. The proposed system can be positioned as an assistive drafting and information-synthesis tool, not an autonomous doctor. Microsoft frames MAIRA as technology intended to assist radiologists, and Google explicitly warns that MedGemma outputs require validation and are not intended to directly determine diagnosis or treatment. [1][6]

Potential impact	How the system could help	Main caveat
Report drafting	Create a first-pass structured report for clinician review.	Draft can contain omissions or false findings.
Workflow speed	Reduce repetitive documentation effort.	Any speed gain is meaningless if error-review burden rises.
Consistency	Standardize section structure and terminology.	Templates can also force the wrong assumptions.
Education	Provide explanations and visual grounding for learning.	Educational output should not be mistaken for clinical advice.
Low-resource settings	Potentially improve access to preliminary interpretation/reporting support.	Validation across populations and imaging devices is essential.
Longitudinal review	Summarize changes across prior studies.	Requires reliable study matching and temporal reasoning.

The most defensible claim for a student prototype is "AI-assisted report drafting / clinical decision support prototype", not "AI diagnosis" or "doctor replacement."

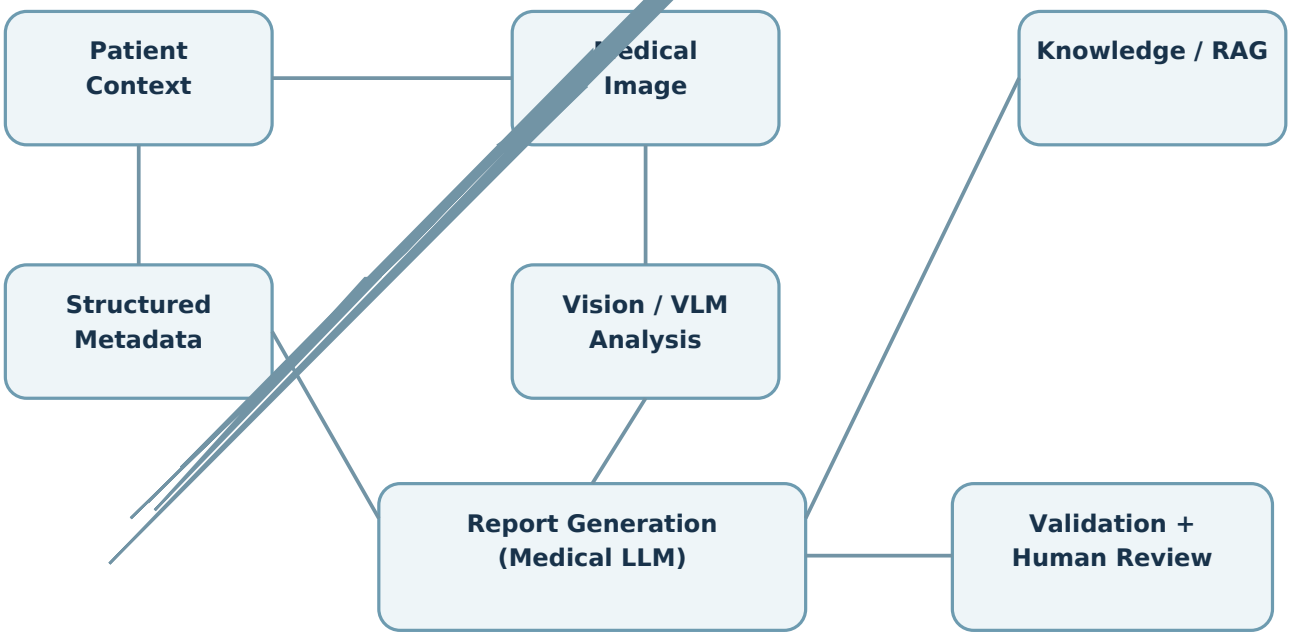
8. Safety, Privacy and Regulatory Considerations

Medical AI has a different risk profile from ordinary consumer AI. WHO guidance highlights ethical and governance concerns around AI in health, while the U.S. FDA has specific guidance addressing clinical decision-support software and when software functions may fall within device oversight. The exact regulatory status depends on intended use, claims, jurisdiction, and implementation. [15][16]

- Hallucination: the model may write fluent but unsupported findings.
- Text bias: multimodal models can be heavily influenced by accompanying text; research has shown that incorrect textual hints can reduce image-based performance. [7]
- Dataset shift: performance can change across hospitals, demographics, scanners, acquisition protocols and prevalence patterns.
- Automation bias: users may over-trust an AI report simply because it looks professional.
- Privacy: medical images and clinical details may contain protected or identifying information.
- Traceability: keep model version, prompt/configuration, evidence sources, and report history for reproducibility.
- Human-in-the-loop: the UI should make clinical review explicit and prominent.

A particularly useful product feature is an evidence panel: every important generated finding should have an associated source such as an image region, structured label, retrieved knowledge passage, or explicit "not available" marker.

9. Recommended System Architecture for a 3rd-Year Project



Recommended MVP flow

Step	Implementation choice
1. Upload	React form + image validation + basic patient context.
2. Preprocess	Resize/normalize; optionally read DICOM metadata.
3. Visual inference	Use a pretrained medical VLM/model; do not train a foundation model from scratch.
4. Context fusion	Combine image findings with symptom/history fields using a structured schema.
5. Retrieval (optional but valuable)	Retrieve relevant reference material or prior report examples to ground terminology and structure.
6. Generate	Generate Findings, Impression, and carefully qualified recommendations in JSON/schema-first format.
7. Verify	Run rule checks / clinical label checks / factuality verifier and show uncertainty.
8. Review	Clinician-style review screen; user can edit/approve the draft.
9. Export	PDF and saved report record.

Suggested stack

Frontend: React + Tailwind | Backend: FastAPI | ML: PyTorch + MONAI where training is required | Image: pydicom/OpenCV/PIL | Database: PostgreSQL | Storage: object storage | Deployment: Docker | AI: pretrained medical multimodal model + optional RAG/verification.

10. What Should Be the Project's Novelty?

Because automatic radiology report generation already has substantial research history, your mentor will probably need a defensible project contribution. The following are realistic options, ordered roughly from easiest to hardest.

Novelty option	Difficulty	Value
Context-aware report generation	Medium	Compare image-only vs image + symptoms/history.
RAG-grounded reports	Medium	Measure whether retrieval improves factuality or reduces unsupported claims.
Evidence-grounded UI	Medium	Show image regions, labels, retrieved evidence and uncertainty.
Factuality verifier	Medium-high	Second model/rule layer checks report against extracted findings.
Longitudinal report generation	High	Compare current and prior X-rays for changes over time.
Multi-modal patient report	High	Fuse image + report + symptoms + lab values in a consistent schema.
Cross-dataset robustness	High	Train on one dataset and test on another to study generalization.

My recommendation: choose context-aware + grounded report generation as the main research question. It is strong enough to be meaningful, but still feasible for an undergraduate team.

11. Team Knowledge Requirements

Knowledge area	What the team needs to know	Priority
Python + backend	APIs, async requests, file handling, validation, auth, logging	Must
Computer vision	Image preprocessing, CNNs/ViTs, transfer learning, inference	Must
Medical imaging	X-ray basics, DICOM, study/view terminology	Must
LLMs/VLMs	Prompting, multimodal input, structured outputs, context windows	Must
Evaluation	Classification metrics, factuality metrics, test splits, human review	Must
RAG	Embeddings, retrieval, chunking, source attribution	Useful
Security/privacy	De-identification, access control, secrets, auditability	Must
MLOps	Docker, model serving, versioning, reproducible experiments	Useful

For a student team, do not require everyone to know every topic. Split ownership across frontend/backend, vision/data, LLM/RAG/evaluation, and deployment/testing.

12. Realistic Scope and Timeline

A strong semester-scale project can be completed without training a large medical foundation model from scratch. The exact timeline depends on team size, GPU access, mentor expectations, and how much training you do.



Illustrative 10-week execution path; actual duration depends on team size and mentor expectations.

Phase	Deliverable
Weeks 1-2	Finalize clinical scope; obtain dataset access; reproduce a baseline; define report schema and evaluation plan.
Weeks 3-4	Image ingestion + pretrained medical model inference + baseline report generation.
Weeks 5-6	Add structured patient context; build backend and report UI.
Weeks 7-8	Add RAG/grounding or factuality verification; build evidence panel.
Weeks 9-10	Evaluation, robustness checks, PDF export, documentation, demo deployment.

What not to put in the MVP: all modalities, autonomous diagnosis, custom LLM pretraining, hospital-grade EHR integration, or a full regulatory-compliance claim.

13. Risks and Mitigations

Risk	Likelihood	Mitigation
Hallucinated findings	High	Grounded schema, evidence display, verifier, abstention and clinician review.
Poor model accuracy	Medium-high	Use pretrained medical models, restrict modality, evaluate clinically relevant labels.
Dataset access delays	Medium	Start with accessible benchmark samples / IU X-Ray while credentials are pending.
GPU limitations	Medium	Use small/efficient pretrained models; inference-first MVP; cloud GPU only when necessary.
Scope explosion	High	Chest X-ray first; defer CT/MRI and longitudinal features.
Privacy mistake	High	Use de-identified public data; never expose real patient data in demos without approval.
Weak novelty	High	Define a measurable hypothesis: context/RAG/grounding improves factuality or usability.

14. Recommended Final Project Definition

A precise title that better reflects a strong student implementation could be:

"Context-Aware Grounded Medical Report Generation from Chest X-Rays Using Multimodal AI"

Proposed problem statement: Design and develop an AI-assisted system that accepts a chest X-ray and relevant patient context, extracts image-grounded findings using a pretrained medical multimodal model, optionally retrieves supporting knowledge, and generates a structured radiology-style report with explicit evidence/uncertainty and human review.

Core hypothesis

Adding structured clinical context and evidence grounding to a pretrained multimodal model can improve the usefulness and factuality of generated reports compared with an image-only report-generation baseline.

Primary research experiment

Compare:
A. image-only report generation
B. image + patient context
C. image + patient context + retrieval/grounding

Evaluate on clinical content, factuality, readability, and uncertainty. This creates a clear experimental story for a report, poster, or paper.

Bottom line

This is a strong 3rd-year project if you build an assistive, scoped system around existing medical AI rather than attempting to create a general-purpose medical doctor. The project is technically rich, has real-world relevance, and has a clear research path - but it needs disciplined scope and serious evaluation.

15. Key References and Links

[1] Microsoft Research, Project MAIRA - Multimodal AI for Radiology Applications. https://www.microsoft.com/en-us/research/project/project-maira/

[2] Bannur et al., "MAIRA-2: Grounded Radiology Report Generation," 2024. https://arxiv.org/abs/2406.04449

[3] Ranjit et al., "Retrieval Augmented Chest X-Ray Report Generation using OpenAI GPT models," PMLR 219, 2023. https://proceedings.mlr.press/v219/ranjit23a.html

[4] Google DeepMind, MedGemma - medical text and image comprehension models, current documentation. https://deepmind.google/models/gemma/medgemma/

[5] Li et al., "LLaVA-Med: Training a Large Language-and-Vision Assistant for Biomedicine in One Day," 2023. https://arxiv.org/abs/2306.00890

[6] Google Developers, Google I/O 2025 keynote recap - MedGemma. https://developers.googleblog.com/en/google-io-2025-developer-keynote-recap/

[7] Buckley et al., "Multimodal Foundation Models Exploit Text to Make Medical Image Predictions," 2023. https://arxiv.org/abs/2311.05591

[8] Chen et al., "Generating Radiology Reports via Memory-driven Transformer (R2Gen)," EMNLP 2020; code. https://github.com/zhjohnchan/R2Gen

[9] Curated collection: "awesome-radiology-report-generation," GitHub. https://github.com/zhjohnchan/awesome-radiology-report-generation

[10] Event-AHU, Medical Image Analysis research/code collection, including recent radiology report-generation work. https://github.com/Event-AHU/Medical_Image_Analysis

[11] PhysioNet, MIMIC-CXR Database v2.1.0. https://physionet.org/content/mimic-cxr/2.1.0/

[12] PhysioNet, MIMIC-CXR-JPG v2.1.0. https://www.physionet.org/content/mimic-cxr-jpg/2.1.0/

[13] Stanford ML Group, CheXpert dataset / publication information. Source summary: Pranav Rajpurkar's Stanford thesis and CheXpert documentation.

[14] Jain et al., "RadGraph: Extracting Clinical Entities and Relations from Radiology Reports," 2021. https://arxiv.org/abs/2106.14463

[15] World Health Organization, "Ethics and governance of artificial intelligence for health," 2021. https://www.who.int/publications/i/item/9789240029200

[16] U.S. FDA, Clinical Decision Support Software - Final Guidance. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-decision-support-software

Research note: This report synthesizes public sources available on or before 23 August 2026. It is intended for project planning and academic research, not clinical guidance. Current model capabilities, licenses, data-access requirements, and regulatory interpretations should be rechecked before implementation or deployment.